## Activity 1.1 – Parameterized ALU

### 1. Design Approach:

Designed a 32-bit parameterized ALU using Verilog HDL. It supports ADD, SUB, AND, OR, XOR, SLT, logical left shift and logical right shift operations. A case statement selects the operation. Four flags are generated: Zero, Negative, Carry and Overflow.

### 2. Verification Steps:

- Created a separate testbench.
- Applied 26 test vectors.
- Tested all eight ALU operations.
- Checked zero, maximum values and overflow conditions.
- Compared actual outputs with expected outputs.
- Simulated using ModelSim.

### 3. Results:

The testbench checks the ALU output and status flags. PASS/FAIL messages are displayed for each test.

### 4. Issues Faced:

Output values and status flags must be checked carefully, especially during overflow and boundary-value operations.

### 5. Conclusion:

Successfully designed and verified a parameterized ALU supporting arithmetic, logical, comparison and shift operations.